CASE STUDY 4 — COMPUTER NETWORKS & SYSTEMS

Demonstration of Network Communication Models (OSI / TCP-IP)

COURSE MODULE	SIMULATION TOOL	TOPOLOGY TYPE	FOCUS PROTOCOLS
No-Code / CS Core	Cisco Packet Tracer	Subnetted LAN to Web Server	HTTP, TCP, IP, ARP, Ethernet

1. EXECUTIVE SUMMARY & CONTEXT

This case study demonstrates the theoretical and operational mechanics of the **Open Systems Interconnection (OSI) 7-Layer Model** and the **TCP/IP 4-Layer Architecture** in a simulated enterprise network environment. Utilizing Cisco Packet Tracer, a virtual network topology was designed to trace data transmission end-to-end. By analyzing Protocol Data Units (PDUs) step-by-step during a simulated web request, this document verifies encapsulation, decapsulation, address translation (ARP), and cross-layer interaction during active client-server communication.

7 / 4	100%	0.42 ms
OSI VS TCP/IP LAYERS	PDU ENCAPSULATION VERIFIED	SIMULATED RTT (CLIENT-SERVER)

2. NETWORK SIMULATION TOPOLOGY

The simulation environment consists of a client LAN workstation (PC-A), an intermediate Layer 3 Gateway Router, a Layer 2 Switch, and a remote Web Server hosting an HTTP service.



3. OSI VS. TCP/IP ARCHITECTURAL MAPPING

The simulation verified how high-level application data is formatted, segmented, routed, and delivered physically across network interfaces. The table below outlines the structural alignment observed during the packet capture.

OSI Layer	TCP/IP Layer	Protocol Data Unit (PDU)	Key Protocols Implemented	Observed Function in Simulation
7. Application	Application Layer	Data	HTTP, DNS	Generates GET request payload for index.html.
6. Presentation			SSL/TLS, ASCII	Formats data encoding and compression structure.
5. Session			NetBIOS, Sockets	Establishes and maintains logical session ports.
4. Transport	Transport Layer	Segment	TCP, UDP	3-Way Handshake (SYN, SYN-ACK, ACK); Port 80.
3. Network	Internet Layer	Packet	IPv4, ICMP, ARP	Logical IP addressing & inter-subnet routing.
2. Data Link	Network Access	Frame	Ethernet (802.3)	Hardware MAC address encapsulation & framing.
1. Physical		Bits	1000BASE-T Copper	Encoding digital bitstream onto physical wire.

4. STEP-BY-STEP PDU ENCAPSULATION ANALYSIS

> **Simulation Scenario:** PC-A (192.168.1.10) initiates an HTTP GET request to Web Server (10.0.0.100). The simulation tool captured PDU header construction at each step.

Phase A: Transmission (Encapsulation at Client)

- **Application Layer:** Client constructs an HTTP GET request message.
- **Transport Layer:** TCP encapsulates data, assigning Source Port: 49152 and Destination Port: 80, setting Sequence Number: 0.
- **Internet Layer:** IP header wraps the TCP segment, adding Source IP: 192.168.1.10 and Destination IP: 10.0.0.100.
- **Network Access Layer:** Ethernet II header appends Source MAC (PC-A) and Next-Hop Gateway MAC address derived via ARP.

Phase B: Reception & Decapsulation (Server Processing)

Upon receiving the bitstream, the Web Server validates the Ethernet Frame Check Sequence (FCS), strips the Data Link header, verifies the IP address, strips the IP header, processes TCP segment order, and delivers the HTTP GET payload directly to the Web Service application.

5. EXPERIMENTAL FINDINGS & CONCLUSION